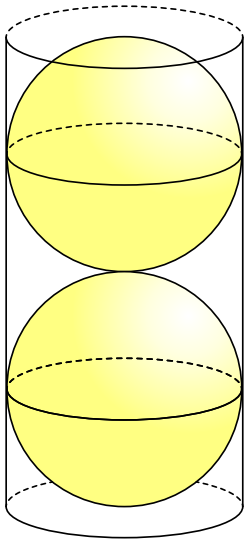
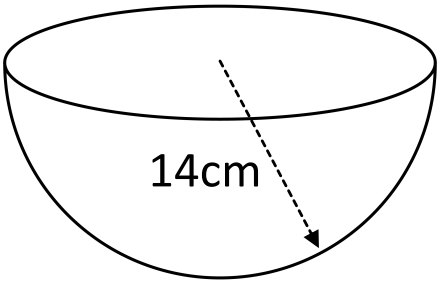
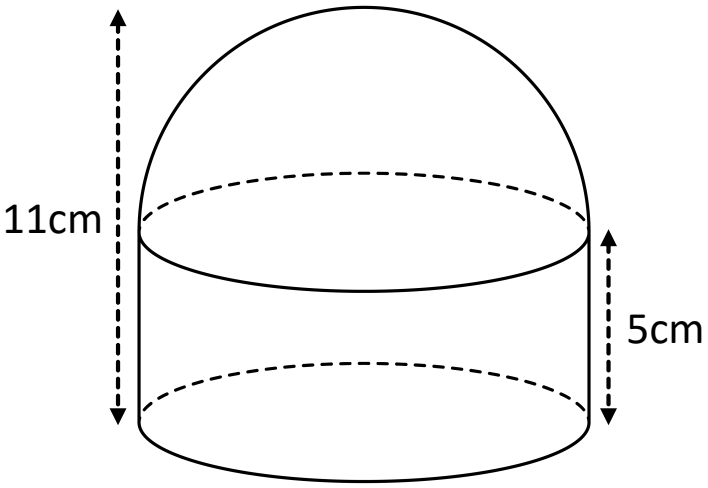
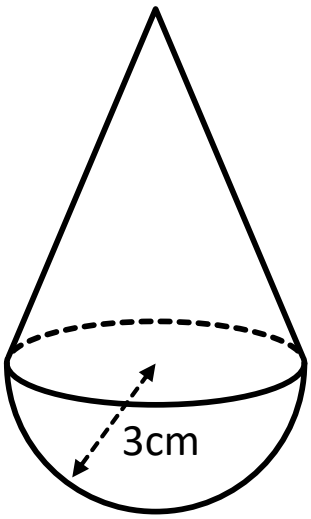
Q1 Craig buys two tennis balls which fit inside a cylindrical container of the same diameter. The tennis ball's diameter is 6cm.What fraction of the container's volume do the tennis balls take up?



Q2 Marie has some hanging flower baskets in the shape of hemispheres with radius 14cm. Marie is going to fill the flower baskets completely with soil. She has 50 litres of compost. **1 litre = 1000cm³**. Work out how many flower baskets Marie can fill completely with compost.



Q3 The following composite solid is made by joining a cylinder and hemisphere. Calculate the volume of the solid leaving your answer in terms of π.



Q4 A child's toy is made from joining a cone and a hemisphere. The cone and the hemisphere both have a radius of 3cm. The ratio of the height of the toy to the radius of the hemisphere is 3:1. Calculate the volume of the toy in terms of π.